

Syllabus

Fall semester 2025

Course Title	Introduction to Earth Science		Course Code		PE1201		Credit	3	Classroom	Tam2403
			Type				Hours	3		
Teacher	Name	Class no.	Office	Office hours	Contact			E-mail		
	Park, Changhyoun							idrpak@pusan.ac.kr		

1. Course Description

This course offers a comprehensive introduction to Earth Science, exploring the dynamic systems that shape our planet. Using *The Blue Planet* by Skinner as the primary textbook, students will investigate Earth's structure, surface processes, atmosphere, oceans, and the complex interactions between human activity and natural systems. Topics include geosphere, hydrosphere, atmosphere, and climate systems and change. Through lectures, discussions, and hands-on activities, students will develop a foundational understanding of Earth as an interconnected system and gain insight into the scientific principles that govern our planet.

2. Core Target Competencies

Subject Core Competencies					
1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression/ Spirit of Adventure
Ability to identify, manage and deal with complex problems; Ability to understand and express themselves in two or more languages; Ability to be the first to come up with and present new ideas; Ability to understand, use and apply digital technology; Ability to develop an open mind about the traditions or values of others; and Ability to develop flexibility, agility and adaptability.					

3. Textbooks and References

A. Textbook: *The blue planet, an introduction to earth system science* (B.J. Skinner and B.W. Murchk, Wiley, 2011)

B. References:

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4. Class Method

A. Flipped learning

B. Presentation

C. Discussion

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5. Evaluation

A. Evaluation Criteria

Presentation	Quiz and Discussion	Midterm exam	Final exam	Attendance and Attitude	Sum
20%	10%	30%	30%	10%	100%

B. Evaluation Method

(1) Presentation

- This class is a kind of (semi-) flipped learning.
- Students will present their research findings to the class, using PPT slides. They will be graded on content, clarity, and presentation skills.

(2) Quiz and Discussion

- Periodic Quiz to assess students' understanding of key concepts and terms. These will be short, covering recent topics. Active participation needs.
- Students are encouraged to ask questions and contribute ideas.

(3) Mid-term exam

- A comprehensive exam that covers the first half of the course material.
- It will include short answer and essay questions.-

(4) Final exam

- A comprehensive exam covering the second half of the course material. The format will be similar to the midterm exam but more extensive.
- The final exam will include short answer and essay questions.

(5) Attendance & Attitude

- The standard of failure of class is followed by the school's regulation.
- However, one point deduction for one missing class.
- Active participation in class discussions, group activities, and engagement with course materials.

6. Criteria for Awarding Credits

☞ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

- Attendance for at least three-quarters of the total class hours is required to earn credit.
 - For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
 - Calculation of the basis score for awarding credit to absentees: **A deduction of 1 point for each absence.**
 - If a basis score cannot be calculated, the grading method : **Assigned an F for the performance.**
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 8.18.~8.22.	The Earth system (1)	
Core Competencies	◆ Earth system	
Week 2 8.25.~8.29.	The Earth system (2)	
Core Competencies	◆ matter	
Week 3 9.1.~9.5.	The Earth system (3)	
Core Competencies	◆ energy	
Week 4 9.8.~9.12.	The Earth system (4)	
Core Competencies	◆ the sun and the earth	
Week 5 9.15.~9.19.	The atmosphere (1)	
Core Competencies	◆ the atmosphere	
Week 6 9.22.~9.26.	The atmosphere (2)	
Core Competencies	◆ wind and weather system	
Week 7 9.29.~10.3.	The atmosphere (3)	National Foundation Day (10.03.)
Core Competencies	◆ the climate system (1)	
Week 8 10.6.~10.10.	The atmosphere (4)	
Core Competencies	the climate system (1)	

Week 9 10.13.~10.17.	Midterm Exams (10.13.~10.16.)	10.17. Regular classes
Week 10 10.20.~10.24.	The changing Earth system	An invited talk
Core Competencies	♦	
Week 11 10.27.~10.31.	The hydrosphere (1)	
Core Competencies	♦ The frozen hydrosphere	
Week 12 11.3.~11.7.	The hydrosphere (2)	
Core Competencies	♦ The world ocean	
Week 13 11.10.~11.14.	The hydrosphere (3)	
Core Competencies	♦ The world ocean	
Week 14 11.17.~11.21.	The geosphere (1)	
Core Competencies	♦ The dynamic geosphere	
Week 15 11.24.~11.28.	The geosphere (2)	
Core Competencies	♦ Earthquakes and volcanoes	
Week 16 12.1.~12.5.	Final Review	Make-up class week
Week 17 12.8.~12.12.	Final Exams (12.08.~12.12.)	

※ The above plans may change according to the academic schedule.